

Rishi Bala

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Education

The University of Texas at Austin, Austin, Texas

August 2025 – May 2029

- GPA: 3.94/4, BS: Electrical and Computer Engineering Honors, BBA: Canfield Business Honors, ACT: 36/36
- Relevant Coursework: Digital System Design Using HDL, Digital Logic Design, Circuit Theory, Introduction to Embedded Systems

Experience

Longhorn Racing – Electric | Austin, Texas

August 2025 – Present

Electronics Engineer (Competing in the Formula SAE International Competition)

- Designed the car's thermal management system, implemented closed-loop fan control and flow rate regulation to optimize battery usage.
- Wrote latency-optimized firmware for real-time data acquisition and live CAN-based telemetry across all vehicle subsystems.
- Built and planned an automotive-grade wiring harness, including connector selection, crimping, and full vehicle integration.

Competitive VEX Robotics | Summit, New Jersey

September 2021 – May 2025

Captain, Lead Builder, Lead [Designer/Documenter](#)

- Captain of a six person VEX team; spent 20+ hours each week holding team meetings for our robot and mentoring younger students.
- Employed CAD software such as Autodesk Fusion 360; documented each robot iteration in an award-winning engineering notebook.
- 1x World Finalist, 1x Worlds Division Champion, 1x Worlds Division Finalist, 2x New Jersey Tournament Champion

New Jersey Institute of Technology Intern | Physics and Mechanical Engineering Laboratory

June 2024 – August 2024

Intern – Prof. Nuggehalli Ravindra and Prof. Balraj Mani

- Researched and worked on two review papers detailing [magnetotactic bacteria \(MTB\)](#) and ferrofluids as a primary author.
- Utilized CAD to design and produce an array of milli-scale electromagnetic coils driven by H-bridges to control miniature robotic arms.

VEX IQ Instructor | Summit, New Jersey

November 2021 – Present

Teacher, Judge, Referee

- Taught VEX IQ students throughout the year, introducing students to basic robotics techniques and block coding.

Projects | www.rishibala.com

Thermal Sensor Module (TSM)

August 2025 – Present

[Github Repository](#) | [Hardware Pictures](#)

- Led the development of the TSM, an enclosure combining radiator fan control with integrated cooling loop and flow rate monitoring.
- Owned the entire design process, including component selection, schematic design, soldering, testing, and firmware development.
- Wrote latency-optimized firmware using FreeRTOS on top of an STM32 microcontroller; developed a CAN-based telemetry system.

Embedded Shuffleboard

March 2026 – Present

[Github Repository](#) | [Hardware Pictures](#)

- In charge of designing a multiplayer embedded shuffleboard platform as part of the ECE319H (Embedded Systems) final project.
- Implemented mixed signal I/O, comparator logic with multiplexer selection, visual/auditory feedback, and a 8-segment display driver.
- Wrote embedded firmware (C) to handle collision physics, input devices, and SPI communication with the LCD, and speaker/amplifier.

YerraGlasses

December 2025 – March 2026

[Github Repository](#) | [Hardware Pictures](#)

- Designed audio-augmented smart glasses using an ESP32-S3 and dual I2S Class-D audio amplifiers for real-time auditory feedback.
- Developed embedded machine learning leveraging neural processing unit object detection to convert visual input into real-time audio.
- Created a custom space-constrained PCB and power system with USB-C, rail regulation, and I2S audio for low-latency operation.

6502 Breadboard Computer

August 2024 – December 2024

- Built a breadboard computer using the classic 6502 microprocessor; programmed simple instructions in assembly.
- Created a custom clock module using a separate Arduino Uno; used an Arduino Mega to communicate with the 6502 computer.
- Utilized an EEPROM chip to store machine code and communicate with the processor; used several LCDs and an array of electronics.

Additional Information and Skills

Technical Skills: C, Java, Python, JavaScript, MongoDB, Node.js, CAD, RTOS, PCB Design

Honors: VEX Robotics World Finalist, World Division Champion, National Merit Commended Scholar

Interests: Pickleball, Motorsport, Drums, Jazz, Tennis, Go-Karting

Work Eligibility: Eligible to work in the U.S. with no restrictions